# Linguistics in the Age of Data: Introspection vs. Observation

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# Outline


- How Linguists Collect Data
  Well-formedness in Linguistics
- The Data Desert of the 1950s
  - Harris's Distributional Program
  - The Methodological Impasse
- Chomsky's Revolutionary Pivot
- A Little History: Two Cultures
- Corpora and Structural Discovery
- The LLM Revolution (2017–2025)
  - Scaling Laws and Emergence
  - GPT-4, Claude, Llama, Gemini
  - RLHF and Alignment
  - Reasoning Models
- Revisiting Method and Theory

# Two Cultures of Linguistics

**Rationalist Tradition**

- Language as innate mental faculty
- Grammar discovered through introspection
- Grammaticality judgments as primary data
- Competence over performance
- Chomsky, generative grammar

**Empiricist Tradition**

- Language learned from experience
- Structure discovered from observed usage
- Distributional patterns as primary data
- Performance reveals competence
- Harris, Firth, corpus linguistics

**The central question:** Where does linguistic knowledge come from?

# The Thesis of This Lecture

**Main Argument**

The **rationalist/introspective turn** in linguistics (Chomsky, 1957) was not merely a theoretical preference—it was a **methodological necessity** forced by the **absence of data** required for distributional analysis.

**Key claims:**

1. Harris's distributional program (1951) was methodologically sound
2. But the data required to implement it did not exist
3. Chomsky's introspective method was a brilliant workaround
4. The empiricist program has been vindicated by modern NLP

*"Harris was right about the method—but 60 years too early."*

# Investigate Hypotheses by Consulting Native Speakers' Intuitions

- Most linguists assume that people can distinguish strings of words that are sentences of their language from strings of words that are not sentences of their language.
- So imagine that you are a machine or a classifier that takes a sentence as input, and returns accept or reject as output.

# Speakers as Automata: Accept/Reject Strings of Words

- ✓ The student read a book.
- × Student the a read book.
- × Of to dog every walks.
- ✓ Every dog walks fast.

# Grammaticality

- A string of words that you recognize as a sentence in your native language is grammatical.
- A string of words that you do not recognize as a sentence in your native language is ungrammatical.
- When you decide whether a sentence is grammatical or ungrammatical, this is called giving a grammaticality judgment.
- Ungrammatical sentences are preceded by an asterisk or star (*). Sometimes they are called starred sentences.

# Grammatical ≠ Meaningful

- 1. It is unlikely that Pat will succeed.
  2. It is improbable that Pat will succeed.
  3. Pat is unlikely to succeed.
  4. *Pat is improbable to succeed.
  This could be meaningful, but most people consider it to be ungrammatical
- 1. They saw Pat with Chris.
  2. They saw Pat and Chris.
  3. Who did they see Pat with?
  4. *Who did they see Pat and?
  Again, this could be meaningful, but it is ungrammatical.

## The Era of Fieldwork

# Part I.A

## The Americanist Tradition

How Linguists Collected Data Before the Computer

# The "Boasian" Mandate (1900–1950)

Before we talk about the "Data Desert," we must acknowledge the "Data Pioneers."

**The Context: American Structuralism**

- Figures: Franz Boas, Edward Sapir, Leonard Bloomfield.
- **The Crisis:** Indigenous languages of the Americas were vanishing.
- **The Goal:** "Salvage Linguistics" — documenting languages before they were lost.

**The Difference**

Modern NLP scrapes *existing* data (web, books).
The Americanists had to **create** the data from silence.

# The Methodology: Fieldwork

**The Setup:**

- The linguist travels to the community (or the speaker comes to the university).
- **Face-to-face interaction** with a native speaker ("informant" or "consultant").
- **The Medium:** Pen, paper, and ears.

**Three Modes of Collection:**

1. **Word Lists:** Collecting basic vocabulary (Swadesh lists).
2. **Paradigms:** Eliciting grammatical forms (“I go”, “You go”, “We go”).
3. **The Text:** The narrative (myths, stories) dictated slowly for transcription.

# The Technology of Observation

**The Notebook**

- Phonetic transcription (IPA or Americanist notation).
- Interlinear glossing (word-for-word translation).
- *The primary database.*

**Audio Capture**

- **1900s:** Wax Cylinders (2 mins, fragile).
- **1930s:** Aluminum discs.
- **1940s: Wire Recorders** (magnetic wire).
- **1950s:** Magnetic Tape (Reel-to-Reel).

*Note: Audio was for verification. Analysis was still done manually on paper.*

# Beyond the "Founding Fathers"

**The Boasian Circle was not a boys' club.**

While Boas, Sapir, and Bloomfield formalized the *theory*, much of the **data collection** and methodological innovation was driven by women.

**Key Figures in Data Collection:**

- **Zora Neale Hurston:** Folklore and the "Insider" Method.
- **Ella Deloria:** The Native Linguist/Analyst.
- **Frances Densmore:** The Pioneer of Audio.
- **Mary Haas:** Institutionalizing Field Methods.
- **Margaret Mead:** Visual Anthropology and patterns.

# Zora Neale Hurston: The Insider Method

**Refining the "Narrative Text"**

- A student of Boas at Columbia (1920s).
- collected African American folklore in the South and Caribbean (*Mules and Men*).
- **Methodological Innovation:** She rejected the "detached observer" model. She "embedded" herself to collect **naturalistic discourse** rather than stiff, elicited sentences.



*"Research is formalized curiosity. It is poking and prying with a purpose."*

# Ella Deloria: Breaking the "Informant" Binary

**The Yankton Dakota Linguist (1889–1971)**

Standard field methods relied on a hierarchy:

$$\textit{White Linguist (Analyst)} \leftrightarrow \textit{Native Speaker (Informant)}$$

**Deloria broke this model.**

- She was Boas's co-author on the *Dakota Grammar* (1941).
- **Methodological Superiority:** Because she was a native speaker *and* a trained linguist, she could verify data via **introspection**.
- She anticipated the Chomskyan focus on "native speaker intuition" by 30 years, correcting Boas's errors in the field.

# Frances Densmore: The Technology of Capture

**Preserving the Signal**

- Ethnomusicologist for the Smithsonian (BAE).
- Traveled with heavy wax cylinder phonographs.
- Recorded over 2,500 cylinders of Indigenous music and speech.

**Why this matters for Linguistics:**

- She proved that **prosody, tone, and rhythm** were essential data points, not just the written segment.
- Her archives remain a primary data source for language revitalization today.

# Mary Haas: Institutionalizing the Method

**The Bridge to Modernity**

- Student of Sapir; arguably the most prolific fieldworker of the mid-century.
- Worked on Tunica, Natchez, Creek, and Thai.
- **The Berkeley School:** She founded the *Survey of California Indian Languages* at UC Berkeley.

**Her Contribution:**

- She turned "Field Methods" from an art into a **teachable curriculum**.
- The strict protocols of notebooks, slips, and file management that defined the 1950s "Shoebox" era were largely standardized by her students.

# Summary: Data is Human

**Correcting the Narrative**

When we talk about the "Data Desert" of the 1950s, we are talking about **computational** sparsity.

But the **ethnographic** record was rich, thanks to:

- **Hurston's** naturalism.
- **Deloria's** intuition.
- **Densmore's** recordings.
- **Haas's** rigor.

*Structuralism wasn't just abstract algebra; it was built on this fieldwork.*

# The "Search Engine" of the 1940s

**The Slip File (The Shoebox)**

How do you analyze distribution without ‘grep‘ or Python?

- **Procedure:**
  1. Copy every word/sentence onto a $3 \times 5$ inch index card ("slip").
  2. File them in a shoebox.
  3. Physically sort them to find patterns.
- **Example:** To study the suffix *-ing*, you physically pull out every card containing *-ing* and lay them on a table.

**This was the "Corpus" of the structuralists.**

# Sample Field Data (Bloomfield/Sapir Style)

**The Interlinear Gloss**

| | | | | |
|---|---|---|---|---|
| *k-im-a:d-ad* | *na* | *mas* | *an-i* | ... |
| 1SG-see-PST-3OBJ | DET | man | who-SG | ... |
| "I saw" | "the" | "man" | "who" | ... |

**The Process:**

1. **Phone:** Transcribe sounds accurately.
2. **Phoneme:** Group sounds into contrastive units.
3. **Morpheme:** Segment words into meaningful parts.
4. **Syntax:** Describe the arrangement of morphemes.

Strictly Inductive: No *a priori* grammar. The grammar emerges from the data.

# Zellig Harris's Role

**Connecting Fieldwork to Formalism**

Harris's *Methods in Structural Linguistics* (1951) was not created in a vacuum.

- It was an attempt to **formalize the fieldwork workflow**.
- **The Goal:** If you follow these procedures (segmentation, substitution), you *must* arrive at the correct grammar.
- **The "Discovery Procedure":** An algorithm for linguistics.

*"The methods... are designed to eliminate the reliance on intuition and to make linguistic analysis replicable."*

# The Problem of Scale

**Why this wasn't enough for Distributional Semantics**

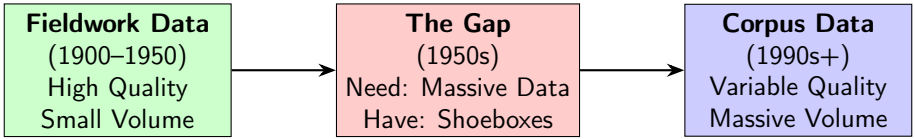
The Boasian method worked for **Morphology** and **Phonology**.

- You can find the phonemes of a language with a few thousand words.

It failed for **Syntax** and **Semantics**.

- Harris's vision: "Difference of meaning correlates with difference of distribution."
- To see distributional differences between synonyms (e.g., *oculist* vs. *eye-doctor*), you need **millions** of words of context.
- You cannot elicit millions of sentences from an informant.
- You cannot store millions of contexts in shoeboxes.

# The Transition




**The irony:** Harris developed the *theory* for the blue box (Corpus Data) while living in the technological reality of the green box (Fieldwork Data).

# Conclusion of This Section

**The "Data Desert" was specific.**

It wasn't that linguists didn't care about data.

- They were obsessed with data! (Field notes, recordings, slips).

The problem was the **type** and **volume** of data.

- **Boas/Sapir/Bloomfield:** Manual analysis of small datasets.
- **Harris's Dream:** Statistical analysis of massive datasets.
- **Chomsky's Pivot:** Since we can't do Harris's dream, let's look inside the mind instead.

# Part I.B

## The Data That Wasn't There

The 1950s: A Methodological Impasse

# Harris's Vision: *Methods in Structural Linguistics* (1951)

**Zellig Harris** laid out a rigorous program:

> *"The danger of using such undefined and intuitive criteria as pattern, symbol, and logical a prioris, is that linguistics is precisely the one empirical field which may enable us to derive definitions of these intuitive fundamental relationships out of correlations of observable phenomena."*
>
> —*Harris and Swanson (1940)*

**The distributional method:**

- Linguistic categories defined by substitutability
- Meaning correlates with distributional similarity
- Grammar discovered from patterns, not assumed *a priori*

**This is exactly what word2vec and transformers do today!**

# The Distributional Hypothesis

Harris (1954), "Distributional Structure"

"Difference of meaning correlates with difference of distribution."

Firth (1957)

"You shall know a word by the company it keeps."

**The insight:**

- Words that appear in similar contexts have similar meanings
- Syntactic categories are defined by distributional equivalence
- No need for intuitions about "noun-ness" or "verb-ness"

**But this method requires one thing: massive amounts of data.**

# The Data That Wasn't There

**What Harris's method required:**

- Large collections of naturally-occurring text
- Ability to count co-occurrences at scale
- Statistical analysis of distributional patterns

**What existed in 1951:**

- No electronic corpora (Brown Corpus: 1961)
- No computers capable of text processing
- Manual concordancing: months to analyze thousands of sentences
- No statistical NLP—wouldn't exist for 40 years

| Resource | 1951 | 2025 |
|---|---|---|
| Corpus size | $\sim$0 | Trillions of words |
| Processing | Manual | GPU clusters |
| Time for analysis | Months/years | Seconds |

# The Scale of the Problem

**How long would Harrisian analysis take by hand?**

To establish distributional classes for English:

- Need millions of word contexts
- Manual recording: ∼10 contexts/minute
- For 1 million contexts: ∼1,700 hours = 212 working days
- Then: cross-tabulation, pattern analysis, statistical tests...

**Result:** A single distributional study would take longer than a career.

### The Methodological Impasse

Harris's procedures were methodologically sound but practically impossible. You could demonstrate them on toy examples, but you couldn't actually *do* linguistics this way at scale.

# The British Empiricist Tradition

**J.R. Firth** (1890–1960) at SOAS, London:

- Emphasized "context of situation"
- Developed collocational analysis
- Influenced Halliday, systemic-functional linguistics

*"You shall know a word by the company it keeps."* —*Firth (1957)*

**The Firthian program survived** in British linguistics, but couldn't flourish without:

- Electronic corpora
- Computational tools for collocation analysis
- Statistical methods for significance testing

**Firth's insight is now the foundation of all neural NLP.**

# Part II

## Chomsky's Revolutionary Pivot

*Syntactic Structures* (1957) as Methodological Rescue

# The Chomskyan Turn

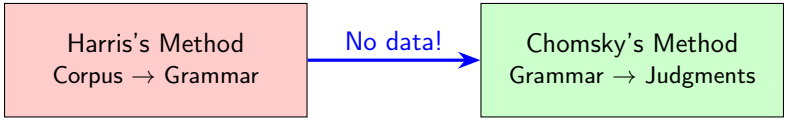
**Noam Chomsky**, Harris's student at Penn, saw the impasse clearly.

*Syntactic Structures* (1957) wasn't just a new theory—it was a methodological rescue operation:

1. From discovery to evaluation
   - Harris: *discover* grammar from distributional data
   - Chomsky: *propose* grammars, evaluate against intuitions
2. Native speaker as oracle
   - A single grammaticality judgment becomes decisive evidence
   - No corpus needed—just a native speaker and clever examples
3. Competence vs. performance
   - Distinguish idealized system from messy usage data
   - *Legitimizes* ignoring distributional data as "performance errors"

# The Brilliant Workaround

**Chomsky's key move:** Replace unavailable corpus data with readily available intuitions.




**Advantages of the new approach:**

- Infinite "data" from a single native speaker
- Can test any hypothesis immediately
- Focus on interesting, revealing examples
- No need for expensive corpus construction

# Poverty of the Stimulus

**Chomsky turned the data problem into an argument:**

**The Poverty of the Stimulus Argument**

Children acquire language from *insufficient* data ⇒ much of grammar must be *innate*.

**The logic:**

1. Children don't get enough data to learn grammar
2. Yet they acquire it perfectly
3. Therefore: innate Universal Grammar

**The rhetorical move:** What was a *practical limitation* (no corpora) becomes a *theoretical principle* (data is insufficient anyway).

If data can't explain acquisition, why bother collecting it?

# *Syntactic Structures* (Chomsky, 1957) 1/3

- *From now on I will consider a language to be a set (finite or infinite) of sentences, each finite in length and constructed out of a finite set of elements.*
- *The fundamental aim in the linguistic analysis of a language L is to separate the grammatical sequences which are the sentences of L from the ungrammatical sequences which are not sentences of L and to study the structure of the grammatical sequences.*
- On what basis do we actually go about separating grammatical sequences from ungrammatical sequences?

# *Syntactic Structures* (Chomsky, 1957) 2/3

- First, it is obvious that the set of grammatical sentences cannot be identified with any finite and somewhat accidental corpus of observed utterances.
- Second, the notion "grammatical" cannot be identified with "meaningful" or "significant" in any semantic sense.
- Sentences (1) and (2) are equally nonsensical, but any speaker of English will recognize that only the former is grammatical.
  1. *Colorless green ideas sleep furiously.*
  2. *Furiously sleep ideas green colorless.*

# *Syntactic Structures* (Chomsky, 1957) 3/3

- Third, the notion "grammatical in English" cannot be identified in any way with the notion "high order of statistical approximation to English." It is fair to assume that neither sentence (1) nor (2) (nor indeed any part of these sentences) has ever occurred in an English discourse.
- Hence, in any statistical model for grammaticalness, these sentences will be ruled out on identical grounds as equally 'remote' from English. Yet (1), though nonsensical, is grammatical, while (2) is not.
- One's ability to produce and recognize grammatical utterances is not based on notions of statistical approximation and the like. I think that we are forced to conclude that grammar is autonomous and independent of meaning, and that probabilistic models give no particular insight into some of the basic problems of syntactic structure.

# The Behaviorist Backdrop

**Collateral damage:** Chomsky's 1959 review of Skinner's *Verbal Behavior*

**The association chain:**

$$\text{Behaviorism} \leftrightarrow \text{Empiricism} \leftrightarrow \text{Corpus linguistics}$$

**Chomsky argued:**

- Skinner's behaviorism couldn't explain language
- Language is too creative, too structured
- Stimulus-response chains are inadequate

**But:** Harris's distributional linguistics $\neq$ Skinner's behaviorism!

The Unfortunate Conflation

Empiricism = behaviorism = wrong.

But distributional linguistics $\neq$ behaviorism!

# The Rationalist Turn

**Chomsky explicitly aligned with:**

- Descartes and innate ideas
- Port-Royal grammarians (17th century)
- Rationalist philosophy of mind

**Against:**

- British empiricists (Locke, Hume)
- American structuralists (Bloomfield, Harris)
- Behavioral psychology (Skinner, Watson)

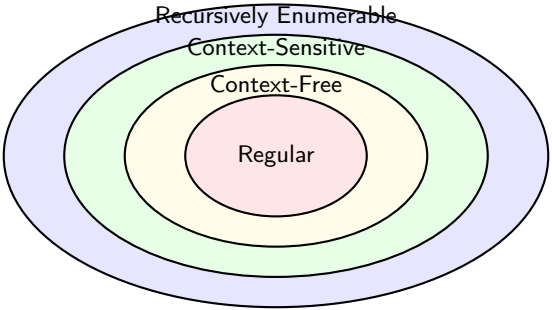
**The Framing**

Language is not learned from experience—it is a window into the innate structure of the human mind.

This framing dominated linguistics for 40+ years.

# Formal Language Theory

**Chomsky's other contribution:** The Chomsky Hierarchy




**Key insight:** You can study the formal properties of grammars without any corpus at all.

- Mathematical linguistics
- Proof-based rather than data-based
- Grammar as algebra

# Foundational Insights: 1940s–1950s

## Automata Theory and Information-Theoretic Models

- Turing's work led first to the McCulloch-Pitts neuron (McCulloch and Pitts, 1943), a simplified model of the neuron as a kind of computing element that could be described in terms of propositional logic.
- Kleene (1951) and (1956) develops models of finite automata and regular expressions.
- Shannon (1948) applied probabilistic models of discrete Markov processes to automata for language.

# Foundational Insights: 1940s–1950s

## Formal Language Theory

- Chomsky (1956), drawing the idea of a finite state Markov process from Shannon's work, first considered finite-state machines as a way to characterize a grammar, and defined a finite-state language as a language generated by a finite-state grammar.
- These early models led to the field of formal language theory, which used algebra and set theory to define formal languages as sequences of symbols.
- This includes the context-free grammar, first defined by Chomsky (1956) for natural languages.

# Foundational Insights: 1940s–1950s

## Noisy Channel Model

- Shannon's metaphor of the noisy channel and decoding for the transmission of language through media like communication channels and speech acoustics.
- Shannon's use of entropy from thermodynamics measures the information capacity of a channel, or the information content of a language; performed the first measure of the entropy of English using probabilistic techniques.
- Sound spectrograph was developed (Koenig et al., 1946), and instrumental phonetics that laid the groundwork for later work in speech recognition.
- First machine speech recognizers in the early 1950s.

# Two Cultures Emerge: 1957–1970

## Speech and Language Processing: Symbolic

- Language Theory: The work of Chomsky and others on formal language theory and generative syntax throughout the late 1950s and early to mid 1960s, and the work of many linguists and computer scientists on parsing algorithms, initially top-down and bottom-up and then via dynamic programming.
- One of the earliest complete parsing systems was Zelig Harris's Transformations and Discourse Analysis Project (TDAP), which was implemented between 1958–1959 at the University of Pennsylvania (Harris, 1962).

# Two Cultures Emerge: 1957–1970

## Speech and Language Processing: Stochastic Paradigm

- Bayesian method was beginning to be applied to the problem of optical character recognition.
- Bledsoe and Browning (1959) built a Bayesian system for text-recognition that used a large dictionary and computed the likelihood of each observed letter sequence given each word in the dictionary by multiplying the likelihoods for each letter.
- Mosteller and Wallace (1964) applied Bayesian methods to the problem of authorship attribution on The Federalist papers.
- The first on-line corpora: the Brown corpus of American English, a 1 million word collection of samples from 500 written texts from different genres (newspaper, novels, non-fiction, academic, etc.), which was assembled at Brown University in 1963–64.

# Rationalism and Empiricism

**Rationalism:**

- The source of knowledge is reason.

**Empiricism:**

- The source of knowledge is data.

**Chomsky** emphasized creativity of language, as manifested in recursive generative rules:
S $\rightarrow$ NP VP, VP $\rightarrow$ VP NP

**Empiricists** emphasize common language patterns (e.g. collocations) and predictability of language:
"About half of the letters or words we choose in writing or speaking (although we are not ordinarily aware of it) are really controlled by the statistical structure of the language." —Warren Weaver (1949)

# Views of Linguistic Data: Rationalism/Internalism

- Language is something in people's minds: a set of rules and principles that allows them to make grammaticality judgments and produce and understand sentences that they have never heard before (i-language or internal language)
- We study i-language by asking people to give grammaticality judgments.
- A corpus (a collection of texts or speech) is e-language, or external language. It is not the object of study.

**Accounts for creativity:** People can produce and understand sentences that have never been uttered before. They are not repeating memorized patterns, but applying productive rules.

# Views of Linguistic Data: Empiricist

- Corpora are the objects of study.
- We study language by examining patterns in corpora (collections of texts or speech).
- Frequency of occurrence in a corpus is easier to measure reliably than a grammaticality judgment.
- Many ungrammatical sentences turn out to be acceptable in the right context.
- Identifying the right context turns out to be an interesting question in its own right.

# Creating a Language Model

Language models are a core component of many applications where the task is to recover an utterance from an input. This is done through Bayes' rule:

$$P(W|O) = \frac{P(O|W) \cdot P(W)}{P(O)}$$

- $W$ = word sequence (what we want to recover)
- $O$ = observation (acoustic signal, corrupted input)
- $P(W)$ — Language Model (prior probability of word sequence)
- $P(O|W)$ — Acoustic/Channel Model (likelihood of observation given words)
- $P(W|O)$ — Posterior probability we want to maximize

# N-gram Language Models

**The Chain Rule:**

$$P(w_1, w_2, \ldots, w_n) = \prod_{i=1}^{n} P(w_i | w_1, \ldots, w_{i-1})$$

**N-gram approximation:** Limit history to previous $n-1$ words

- Bigram: $P(w_i | w_{i-1})$
- Trigram: $P(w_i | w_{i-2}, w_{i-1})$

**Key challenge:** sparsity—most n-grams never observed

# Deconstructing Chomsky's Argument 1/2

Neither of these sentences has ever appeared before in human discourse. None of the substrings has either.

1. *Colorless green ideas sleep furiously.*
2. *Furiously sleep ideas green colorless.*

Hence the probabilities $P(W_1)$ and $P(W_2)$ cannot be relevant to a native English speaker's knowledge of the difference between these sentences.

"*... in any statistical model for grammaticalness, these sentences will be ruled out on identical grounds as equally 'remote' from English.*"

This claim is false: all modern statistical models of language can assign probabilities to previously unseen utterances.

# Deconstructing Chomsky's Argument 2/2

**Language users must be able to generalize beyond their input.**

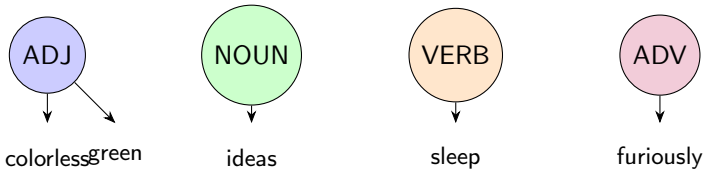
Statistical inference is the study of such generalization.

Pereira (2000) used a class-based bigram model to model (1) and (2):

$$\frac{P(\text{Colorless green ideas sleep furiously})}{P(\text{Furiously sleep ideas green colorless})} \approx 200{,}000$$

Maybe probabilities are not such a useless model of language knowledge after all.

# Class-based Language Models




The $c_i$ are latent variables introduced through annotation or unsupervised learning

$P(w_1 \dots w_n) = \prod_i P(w_i|c_i) \cdot P(c_i|c_{i-1})$

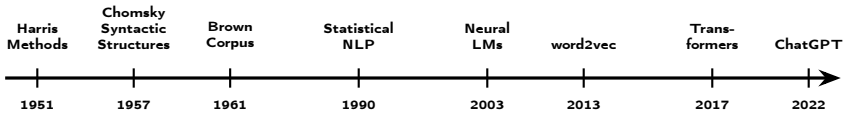
Enables generalization to unseen word sequences

# Part III

## The Empiricist Revival

Data Returns: 1961–Present

# The Timeline




**The pattern:** Distributional methods become viable only when data exists.

- 1961: Brown Corpus (1M words)—first electronic corpus
- 1990s: Penn Treebank, statistical NLP revolution
- 2013: word2vec demonstrates distributional semantics at scale
- 2017–now: Transformers, LLMs—Harris vindicated

# The Brown Corpus (1961–1967)

**Finally, data exists!**

**The Brown Corpus:**

- First standard electronic corpus of English
- 1 million words—enormous for its time
- 500 samples of 2,000 words each
- 15 genres of American English prose
- Cost: $23,000 (1963–64)

**1967:** *Computational Analysis of Present-Day American English*

- First computational corpus linguistics
- Word frequency lists, distributional statistics
- Used for the American Heritage Dictionary (1969)

**One million words.** Today's LLMs train on trillions.

But it took 10 years after Harris for even this modest corpus to exist.

# Empiricism Returns: 1983–1993

- Finite-state models began to receive attention again after work on finite-state phonology and morphology by Kaplan and Kay (1981) and finite-state models of syntax by Church (1980).
- Return of empiricism; most notably the rise of probabilistic models throughout speech and language processing, influenced strongly by the work at the IBM Thomas J. Watson Research Center on probabilistic models of speech recognition.
- These probabilistic methods and other such data-driven approaches spread into part-of-speech tagging, parsing and attachment ambiguities, and connectionist approaches from speech recognition to semantics.

# The Statistical NLP Revolution (1990s)

**Key developments:**

- Penn Treebank (1992): 4.5M words, syntactically annotated
- Statistical parsing (Collins, Charniak)
- Machine translation (IBM Models)
- HMMs for POS tagging

> *"Every time I fire a linguist, the performance of the speech recognizer goes up."* —Frederick Jelinek (IBM), apocryphal

**The paradigm shift:**

- From hand-written rules to learned models
- From introspection to corpus evidence
- From competence to performance data

# Word2Vec (2013): Harris Vindicated

**Mikolov et al. (2013):** Neural word embeddings

**The famous equation:**

$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

**This is pure Harrisian distributional semantics!**

- Words represented by their distributional contexts
- Similarity = co-occurrence patterns
- Meaning emerges from "the company words keep"

**What changed:**

- Billions of words of training data (vs. 0 in 1951)
- GPU computation (vs. manual analysis)
- Neural network optimization (vs. counting)

# Part IV

## The LLM Revolution

2017–2025: Transformers to Reasoning Models

# 2017: Attention Is All You Need

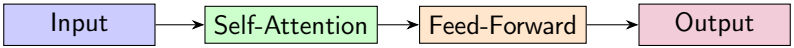
Vaswani et al., Google

The Transformer architecture replaces recurrence with self-attention

**Key innovations:**

- Parallel processing of entire sequences
- Multi-head attention captures different relationship types
- Positional encodings replace sequential processing

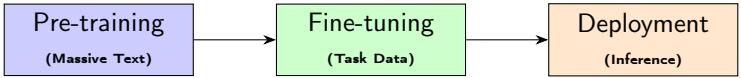
Enabled training on unprecedented scales



# 2018–2019: The Pre-training Revolution

- **BERT** (2018): Bidirectional pre-training via masked language modeling
- **GPT** (2018): Unidirectional autoregressive pre-training
- **GPT-2** (2019): 1.5B parameters, "too dangerous to release"

**Key insight:** Pre-train on massive text, fine-tune on tasks



# GPT-3 (2020): Scale Changes Everything

Brown et al., OpenAI

- 175 billion parameters—100× larger than GPT-2
- Trained on 300B tokens from Common Crawl, books, Wikipedia

**Key finding:** In-context learning emerges at scale

- **Zero-shot**: Perform tasks from instructions alone
- **Few-shot**: Learn from examples in the prompt
- No gradient updates required!

Sparked intense debate: Is this "real" understanding?

# Scaling Laws

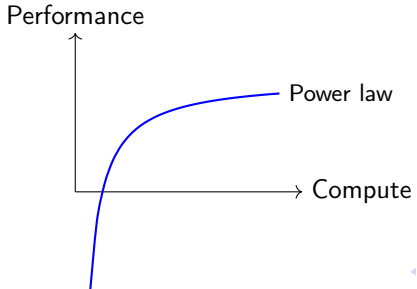
Kaplan et al. (2020), Hoffmann et al. (2022)

**Performance improves predictably with:**

- Model size (parameters)
- Dataset size (tokens)
- Compute (FLOPs)

**Chinchilla scaling (2022):** Optimal ratio is roughly 20 tokens per parameter

**Implication:** We were training models that were too large for their data



# Emergence

Wei et al. (2022)

**Some capabilities appear suddenly at certain scales**

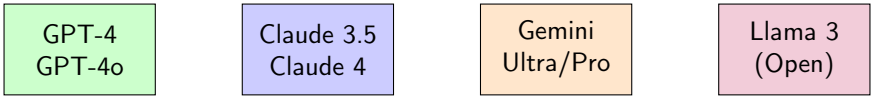
**Examples of emergent abilities:**

- Multi-step arithmetic
- Chain-of-thought reasoning
- Word unscrambling
- Understanding sarcasm

**Controversial:** Is emergence real or an artifact of metrics?

Schaeffer et al. (2023): "Emergence" may reflect metric choice

# The Current Landscape




- **Closed models:** GPT-4, Claude, Gemini—API access only
- **Open weights:** Llama, Mistral, Qwen—downloadable
- Rapid capability improvements across all families

# GPT-4 (2023)

OpenAI

**Multimodal:** Accepts both text and images as input

**Significant improvements in:**

- Reasoning and problem-solving
- Following complex instructions
- Reducing hallucinations
- Professional exam performance (bar exam: 90th percentile)

Architecture details not publicly disclosed

**GPT-4o** (2024): Faster, cheaper, native multimodal
**GPT-4.5** (2025): Further improvements in reasoning and tool use

# Claude (Anthropic)

2023–2025

Founded by former OpenAI researchers (Amodei et al.)

**Focus on AI safety and alignment**
**Constitutional AI:** Train models using AI feedback guided by principles

**Claude 3 family (2024):**

- Haiku (fast), Sonnet (balanced), Opus (capable)

**Claude 3.5 Sonnet:** State-of-the-art on many benchmarks
**Claude 4** (2025): Extended context, improved reasoning

**Notable for:** Long context (200K tokens), coding, analysis

# Gemini (Google DeepMind)

2023–2025

**Google's unified multimodal model family**

**Natively multimodal:** Trained on text, images, audio, video

**Gemini 1.5 Pro:** 1 million token context window

**Integration with Google services:**

- Search, Workspace, Android
- Google AI Studio for developers

**Gemini Ultra:** Flagship model for complex tasks
**Strong on:** Multimodal reasoning, long-context tasks

# Llama (Meta)

## Open-Weight Models

**Meta's strategy:** Release weights openly

**Llama 2** (2023): Up to 70B parameters, commercial use allowed
**Llama 3** (2024): 8B, 70B, 405B variants

- Trained on 15T+ tokens
- Competitive with closed models

**Impact on ecosystem:**

- Fine-tuning and adaptation by community
- Local deployment possible
- Spawned many derivative models

**Philosophical shift:** Open vs. closed development

# The Alignment Problem

**Pre-trained LLMs optimize for next-token prediction**

**But we want models that are:**

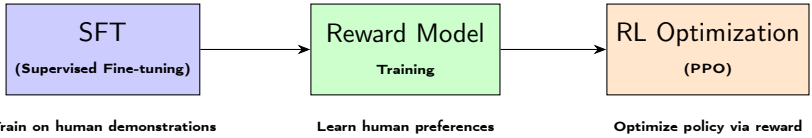
- **Helpful**: Actually answer the user's question
- **Harmless**: Don't produce dangerous content
- **Honest**: Don't make things up

**Gap** between "predict text" and "be a good assistant"

**Solution:** Post-training alignment

# RLHF: Reinforcement Learning from Human Feedback

Ouyang et al. (2022), InstructGPT




- **Stage 1:** Fine-tune on high-quality demonstrations
- **Stage 2:** Train reward model on human comparisons
- **Stage 3:** Optimize LLM to maximize reward

# Constitutional AI

Bai et al. (2022), Anthropic

**Problem with RLHF:** Requires massive human labeling

**Constitutional AI approach:**

1. Define a "constitution" of principles
2. Have the AI critique its own outputs
3. Use AI feedback for training (RLAIF)

**Example principles:**

- "Choose the response that is most helpful"
- "Choose the response that is least harmful"
- "Choose the response that is most honest"

Scales better than pure human feedback

# Direct Preference Optimization (DPO)

Rafailov et al. (2023)

**RLHF is complex:** requires separate reward model + RL

**DPO insight:** Can optimize preferences directly

Treats alignment as a classification problem:

- **Given:** (prompt, good response, bad response)
- **Train:** Increase $P(\text{good})$, decrease $P(\text{bad})$

**Advantages:**

- Simpler training pipeline
- More stable optimization
- No reward model needed

Widely adopted in 2024–2025

# Chain-of-Thought Prompting

Wei et al. (2022)

**Discovery:** LLMs reason better when they "think aloud"

**Chain-of-thought:** Include reasoning steps in output

**Standard:**
Q: Roger has 5 balls. He buys 2 cans of 3. How many balls?
A: 11

**Chain-of-thought:**
Q: Roger has 5 balls...
A: Roger starts with 5. He buys 2 cans $\times$ 3 = 6 balls. Total: $5 + 6 = 11$

# OpenAI o1 and o3 (2024–2025)

## Reasoning Models

**o1** (September 2024): First "reasoning model"

**Key innovation:** Extended internal reasoning before answering

- Uses reinforcement learning to improve reasoning chains

**Performance:**

- 83rd percentile on Codeforces (competitive programming)
- PhD-level performance on science benchmarks
- Major improvements on math olympiad problems

**o3** (December 2024): Further scaling of reasoning

- State-of-the-art on ARC-AGI benchmark
- Significantly improved on mathematical reasoning

# Test-Time Compute Scaling

**Traditional scaling:** More training compute → better models

**New paradigm:** More inference compute → better answers

**Methods:**

- Longer reasoning chains
- Multiple samples + voting
- Search over reasoning paths
- Self-verification and correction

**Trade-off:** Slower but more accurate
**Implication:** Can improve without retraining

# DeepSeek-R1 (2025)

## Open Reasoning Models

**Chinese lab DeepSeek releases open reasoning model**

**Key features:**

- Competitive with o1 on benchmarks
- Open weights available
- Novel training approach published

**Significance:**

- Democratizes access to reasoning capabilities
- Challenges closed-model paradigm
- Enables research into reasoning mechanisms

# Multimodal Foundation Models

**Modern LLMs increasingly multimodal:**

- **Vision**: Understand and generate images
- **Audio**: Speech recognition and synthesis
- **Video**: Temporal visual understanding

**Examples:**

- GPT-4V/GPT-4o: Text + images + audio
- Gemini: Native multimodal training
- Claude: Vision capabilities

**Enables:** Visual question answering, image description, multimodal reasoning

# LLMs as Agents

## Tool Use and Autonomy

**LLMs can now use tools:**

- Web browsing
- Code execution
- API calls
- File manipulation

**Agentic systems:** LLMs that plan and execute multi-step tasks

**Examples:**

- Computer use (Claude)
- Research assistants
- Coding agents (Devin, Cursor)

Raises new questions about autonomy and oversight

# Modern Evaluation Landscape

| Benchmark | Focus | Top Performance |
|---|---|---|
| MMLU | Academic knowledge | >90% |
| HumanEval | Code generation | >90% |
| GSM8K | Math word problems | >95% |
| GPQA | Graduate science | ∼60% |
| ARC-AGI | Novel reasoning | ∼85% (o3) |
| SWE-bench | Real coding tasks | ∼50% |

Many benchmarks now "saturated"
Shift toward harder, more realistic evaluations
Growing concern about benchmark contamination

# The Evaluation Challenge

**Problems with current benchmarks:**

- Test set contamination in training data
- Benchmarks don't capture real-world utility
- Gaming metrics vs. genuine capability

**Emerging approaches:**

- Human evaluation protocols
- Held-out private benchmarks
- Task-specific evaluations
- Red-teaming and adversarial testing

**Open question:** How do we measure "understanding"?

# Part V

## Implications and Conclusions

What Have We Learned?

# What LLMs Tell Us About Language

**LLMs succeed via statistical patterns at massive scale**

**But they exhibit behaviors suggesting more:**

- Compositional generalization
- Abstract reasoning
- Cross-lingual transfer

**The Chomsky question revisited:**

- Can statistical models capture linguistic competence?
- Tentative answer: More than expected, less than hoped

# Chomsky vs. LLMs: The Debate Continues

**Chomsky (2023):** LLMs are "basically a high-tech plagiarism" that tell us nothing about language

**Counter-arguments:**

- LLMs generalize to novel sentences
- They capture grammaticality distinctions
- They show systematic compositional behavior

**Perhaps both sides are partly right:**

- LLMs do capture significant linguistic regularities
- They may not capture the same regularities as humans
- The question of "understanding" remains open

# What the History Teaches Us

1. **Methodology is constrained by technology**
   - Harris's theory was right; his methods were premature
   - Chomsky's methods fit the available tools
2. **Scientific revolutions can be contingent**
   - The "cognitive revolution" in linguistics was partly a response to data scarcity
   - With data, the revolution might have gone differently
3. **Old ideas can be vindicated by new technology**
   - Firth (1957): "You shall know a word by the company it keeps"
   - word2vec (2013): Proves this computationally
4. **The two traditions can be complementary**
   - Rationalist insights about structure and recursion
   - Empiricist methods for learning and discovery

# The Remaining Questions

**What distributional methods *can't* capture (yet):**

- Compositionality and systematicity
- Logical inference and reasoning
- Grounding in perception and action
- Acquisition from limited data (poverty of stimulus)

**What Chomsky got right:**

- Language has recursive, hierarchical structure
- Children learn from surprisingly little data
- Grammar is not just statistical regularities

**The synthesis:**

- Structure + statistics
- Innate biases + learned patterns
- Competence models + performance data

# Implications for This Course

**In CS115B, we work in the empiricist tradition:**

- Learning models from data
- Distributional representations (embeddings)
- Statistical and neural methods

**But we also draw on rationalist insights:**

- Structured representations (trees, graphs)
- Compositional semantics
- Grammar formalisms (CFGs, dependencies)

### The Modern Synthesis

The best NLP combines learned representations with structured computation. Neither pure empiricism nor pure rationalism suffices.

# Summary: The Two Cultures

| | Rationalist | Empiricist |
|---|---|---|
| Data source | Intuitions | Corpora |
| Method | Introspection | Observation |
| Goal | Explain competence | Model performance |
| Grammar | Innate/universal | Learned/statistical |
| Key figures | Chomsky | Harris, Firth |
| Era | 1957–1990 | Pre-1957, 1990–now |
| Modern form | Formal syntax | Neural NLP |

**Key insight:** The shift to rationalism wasn't just theoretical—it was forced by the absence of data for empiricist methods.

**Now that data exists,** empiricism has returned—and vindicated Harris's vision.

# Conclusion

**Linguistics is now both a theoretical and experimental discipline**

The scope of observed data for language study and theorizing is richer and broader than ever

**LLMs have transformed our understanding of what statistical learning can achieve**

**Key developments 2020–2025:**

- Scaling laws and emergent capabilities
- RLHF and alignment techniques
- Reasoning models (o1, o3, DeepSeek-R1)
- Multimodal and agentic systems

The rationalism/empiricism debate continues—but the terms have changed

# Final Thought

*"Linguistics is precisely the one empirical field which may enable us to derive definitions of these intuitive fundamental relationships out of correlations of observable phenomena."*

— Harris and Swanson (1940)

Harris was right.

He just needed to wait 70 years for the data.

# Key References

**Primary sources:**


- Harris, Z. (1951). *Methods in Structural Linguistics*
- Harris, Z. (1954). Distributional Structure. *Word* 10(2-3)
- Firth, J.R. (1957). A Synopsis of Linguistic Theory. *Studies in Linguistic Analysis*
- Chomsky, N. (1957). *Syntactic Structures*
- Chomsky, N. (1959). Review of Skinner's *Verbal Behavior*


**Modern connections:**


- Mikolov et al. (2013). Efficient Estimation of Word Representations
- Vaswani et al. (2017). Attention Is All You Need
- Brunila & LaViolette (2022). What company do words keep? *NAACL*